Discuss

What is the role of the Operating System?

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How?

- IT will be removing it from the college computers in the coming months. You will need to manually "untick" it from the options at home and in college for now.

Role of the Operating System

Keywords: Operating System; OS; Hardware; Software; System software; Virtual Machine; Abstraction; Graphical User Interface; Command Line Interface; Application Programming Interface; Process; Processor; Memory;

From the AQA Specification:

4.6.1.3 System Software

Understand the need for, and functions of the following system software:

- operating systems (OSs)
- utility programs
- libraries
- translators (compiler, assembler, interpreter)

4.6.1.4 Role of an operating system (OS)

Understand that a role of the operating system is to hide the complexities of the hardware.

Know that the OS handles resource management, managing hardware to allocate processors, memories and I/O devices among competing processes.

Key Definition

The role of an operating system is to:

1. Hide the complexities of the hardware from the user.
2. Manage the computer resources which processes compete for.

An operating system is what makes the hardware useable.

1. Hide the complexities of the hardware from the user.

Virtual Machine

The **virtual machine**, or **platform**, that the operating system provides, helps to **hide the complexities of the hardware** behind layers of operating system software.

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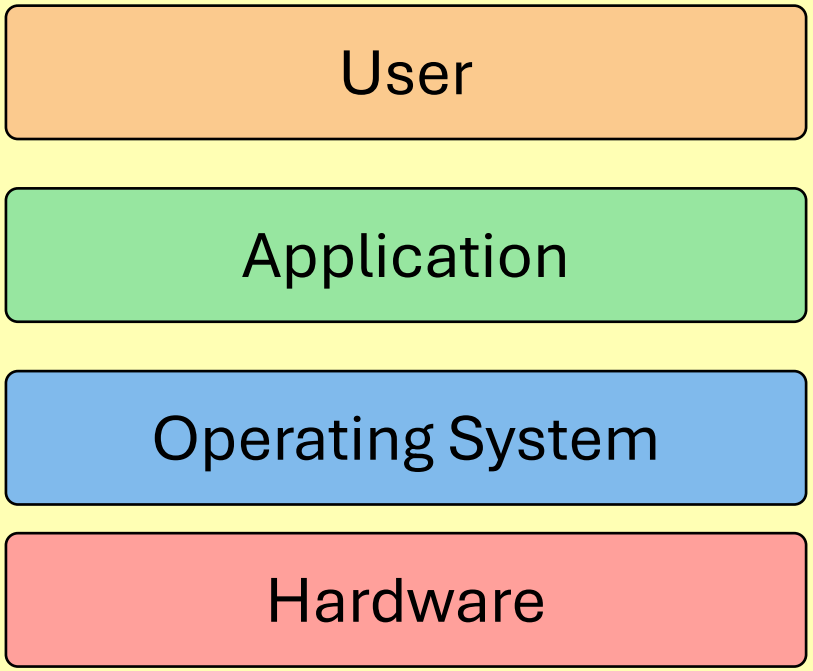
Abstraction!

1. Hide the complexities of the hardware from the user.

The OS provides a platform on which other software can run.
It handles the operations that are common to different pieces of software on the same system.
This allows **applications** to make user interfaces.



How does it do that? (Beyond spec)

1. Interfaces:

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It's the same with MAC & Linux.

How does it do that? (Beyond spec)

2. Application Programming Interface (API)

APIs are layers of software that allow the application programmers to call them to perform common operating system tasks.

eg. opening text files calls a **Windows API function** to execute that command.

2. Manage the computer resources which processes compete for.

What is a process?

Come up with a definition in pairs.

2. Manage the computer resources which processes compete for.

What is a process?

An **instance of a program being executed**. In many modern OS, many processes exist at once.

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Assume we have a single-core processor. There are a number of processes all waiting to be run.
How might the OS decide which one to run next?

Input/Output Devices

The OS will allocate I/O devices to different processes and manage access between processes and devices.

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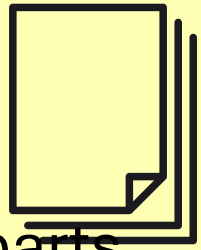
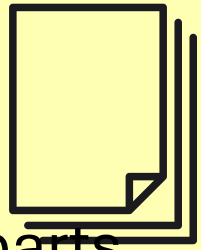
Example: Two processes both want to access the audio input from the microphone.

Memory



The operating system will keep track of and allocate which parts of the main memory are being used by which process.

Memory



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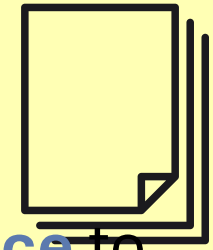
Remember, processes may be allocated blocks of memory that are not **contiguous**.

The OS hides this complexity. Our programs simply don't need to know that most of the time.

Storage



The operating system will allocate **space on a storage device** to files, **organise files into directories**, and determine **where on the device** to save a file.

All of these features are part of a **filesystem**.

Additional functions of the OS (From mark schemes)

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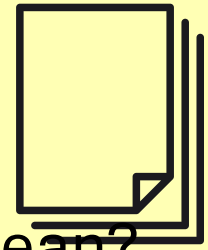
- Handle interrupts
- Installation and automatic updating of software
- Managing power consumption

Dictionaries

Abstract Data Types



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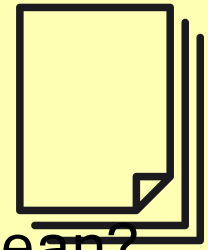
Abstract Data Types



A dictionary is an **abstract data type**, but what does that mean?

An abstract data type is just the rules for how the object **will** function. It doesn't contain algorithms, or even any code.

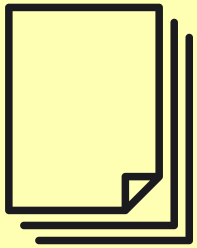
A **data structure** is the realisation of that ADT within a programming language.

Dictionaries

A Dictionary is defined by:

- A collection of (**key**, **value**) pairs.
- Values can only be accessed using the associated key
- There can be <u>no duplicate keys</u>

Dictionaries



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They can be thought of as an array where items are used to access an index, rather than just sequential integers.

A comparison:

Arrays:

```
string[] B007-names =
{"Jim","Sara","Ao"};

B007-names[1] = "Sarah";

Console.WriteLine(
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Dictionaries:

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Dictionary<string,int> B007-scores
          = new Dictionary<string,int>();

B007-scores.Add("Jim",85);
B007-scores.Add("Sara",100);
B007-scores.Add("Ao",40);

B007-scores["Ao"] = 400;

Console.WriteLine(B007-scores["Jim"]);

foreach
    (string name in B007-scores.Keys())
    {
        Console.WriteLine($"Hi {name}!");
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W137 - C# Dictionaries